

Karanjyoti Medhi

GitHub
LeetCode

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EDUCATION

- National Institute of Technology, Silchar** Assam, India
B.Tech in Civil Engineering; CGPA: 7.83
2023 – Present
- Sarala Birla Gyan Jyoti** Guwahati, India
Higher Secondary Certificate (HSC); 86%
2020 – 2022

INTERNSHIP AND HACKATHON EXPERIENCE

- Machine Learning Research Intern, NIT Silchar** Remote
Satyendra Nath Bose Program (Intelligent Traffic Volume Prediction System)
06/2025 – 07/2026
 - Traffic Forecasting Platform:** Developed an end-to-end ML-based traffic volume prediction system using Python, XGBoost, and Scikit-learn on multimodal traffic and weather datasets
 - Feature Engineering & Model Optimization:** Engineered temporal and weather-based features; optimized Regression and Ensemble models using cross-validation and hyperparameter tuning
 - Real-Time Prediction Pipeline:** Built FastAPI-powered REST services and automated ML pipelines for real-time traffic forecasting and intelligent transportation analytics
- Urban Vision Hackathon – IISc Bangalore** Remote
Participant
05/2025 – 06/2025
 - Dataset Development:** Developed large-scale traffic dataset using computer vision annotation, processing 10k+ images for smart city initiatives
 - Team Collaboration:** Collaborated on AI models, improving data accuracy
- 3-in-1 Transformable Bot — SIH Hackathon** Live Webiste — Github(Documentation)
Full Stack Developer & Hardware Integration
09/2025
 - Multi-Modal Robotics:** Engineered transformable drone-boat-car system using PLA biodegradable material with obstacle avoidance VL53L0X sensor
 - Web Platform:** Built educational website with Next.js featuring STEM tutorials and 3D model visualizations
 - System Integration:** Developed wireless control system with NRF24 module implementing mode switching and stunt capabilities

PROJECTS

PORTFOLIO

- Hybrid ATS Resume Reviewer** Live — Github(App) — Github(FastApi)
Next.js, TypeScript, Python, FastAPI, LangChain, NeonDB, Prisma, Redis
08/2025
Built hybrid ATS resume reviewer for on-campus SDE and Data Analyst roles with peer-based rankings
Implemented keyword matching using python REGEX patterns, with GenAI feedback via LangChain + Gemini
Engineered scalable backend with FastAPI + NeonDB via Prisma ORM with OAuth authentication
- Paymenter - Course Marketplace Platform** Live — Github
Next.js, TypeScript, Prisma, NextAuth, Razorpay, MongoDB, Zustand
09/2025
Built a course marketplace with premium subscriptions, secure enrollments, and protected content access
Integrated Razorpay APIs for payment processing, order verification, and subscription billing workflows
Implemented OAuth and credential authentication using NextAuth, Prisma ORM, and MongoDB
- Social Media Upload Platform** Live — GitHub
Next.js 15, TypeScript, React 19, ImageKit, Tailwind CSS
05/2025
Built a full-stack media upload app with ImageKit, supporting 1GB files.
Added bcrypt + NextAuth authentication, and Instagram-style instant background uploads for faster posting.

TECHNICAL SKILLS

- Programming Languages:** Python, C, C++, GoLang, Java, JavaScript
- Frameworks & Libraries:** TypeScript, Node.js, Express.js, React, Next.js, Prisma, Redis, ImageKit, Inngest, LangChain, LangGraph, LangServe, PostgreSQL, MongoDB, NeonDB
- Data Analysis & Machine Learning:** Pandas, NumPy, Matplotlib, Seaborn, Regression, Supervised & Unsupervised Learning, Regular Expressions(Regex)
- Tools & Platforms:** Git, GitHub, Postman, Docker, MongoDB Compass, Jupyter Notebook
- Frontend Technologies:** Tailwind CSS, Zustand, shadcn/ui, Radix UI, WebSockets, WebRTC
- Concepts:** Razorpay integration, OAuth, REST APIs, DB optimization, Responsive design, JWT auth
- Soft Skills:** Leadership, Event Management, Writing, Public Speaking, Teamwork

LEADERSHIP & ACTIVITIES

- **Team Lead, Hackathon:** Led national-level hackathon project, coordinating team of 5 members
- **E-Cell Event Management:** Supported logistics and execution of 10+ events for entrepreneurship cell

CERTIFICATIONS

- **AWS Educate:** Machine Learning Foundations & Introduction to Generative AI
- **Deloitte:** Technology Job Simulation
- **HackerRank:** Software Engineer Intern Certification
- **Urban Vision Hackathon:** IISC Bangalore Participation